

CRESCENDO

Team Name-Crescendo

Sub Domain-

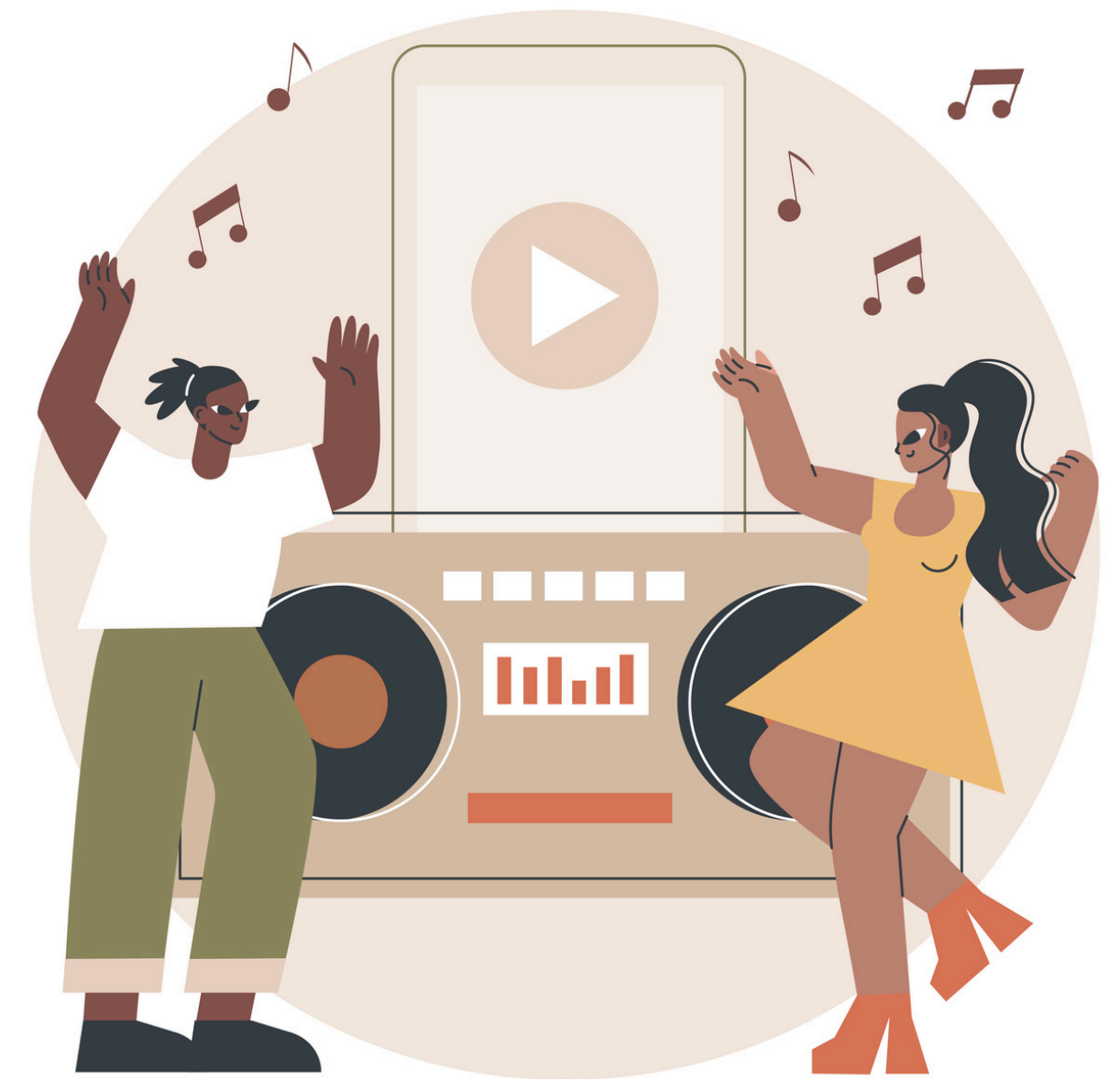
AI for Personalized Music Therapy

Problem statement-

Build a platform that uses AI to recommend music tracks based on user region, user preferences and moods.

Project Category-Software

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CRESCENDO-WEBSITE



Crescendo

Welcome to Crescendo! We help you find the perfect playlist for any occasion. Whether you're looking to create a custom playlist or find the ideal tracks to match your mood, Crescendo has got you covered. Dive into the world of personalized music experiences. At Crescendo, we believe music should adapt to you. Our platform allows you to create playlists tailored to your taste or discover curated selections that reflect your current mood. From energizing anthems to soothing melodies, Crescendo offers a seamless way to match your music to your life. Explore the endless possibilities of music personalization with Crescendo, your ultimate companion in creating the perfect soundtrack for your moments.

Create Playlist

Create a personalized playlist based on your favorite genres, artists, and preferences.

Create playlist

Mood Playlist

Let us suggest the perfect playlist for your mood and provide therapy! Relax, energize, or focus with us!

Explore mood music

Key Features:

Crescendo has two chatbots that help give a customized playlist on various parameters.

- **Create Playlist:** Curates playlists tailored to your activities, enhancing your experience with the perfect music selection.
- **Mood playlist:** Provides mood-based song recommendations and offering therapeutic support, blending music with emotional well-being

Feedback: Users can provide feedback on the recommended playlist, and the AI chatbot will refine the playlist accordingly to better match the user's preferences or mood for a more tailored experience.

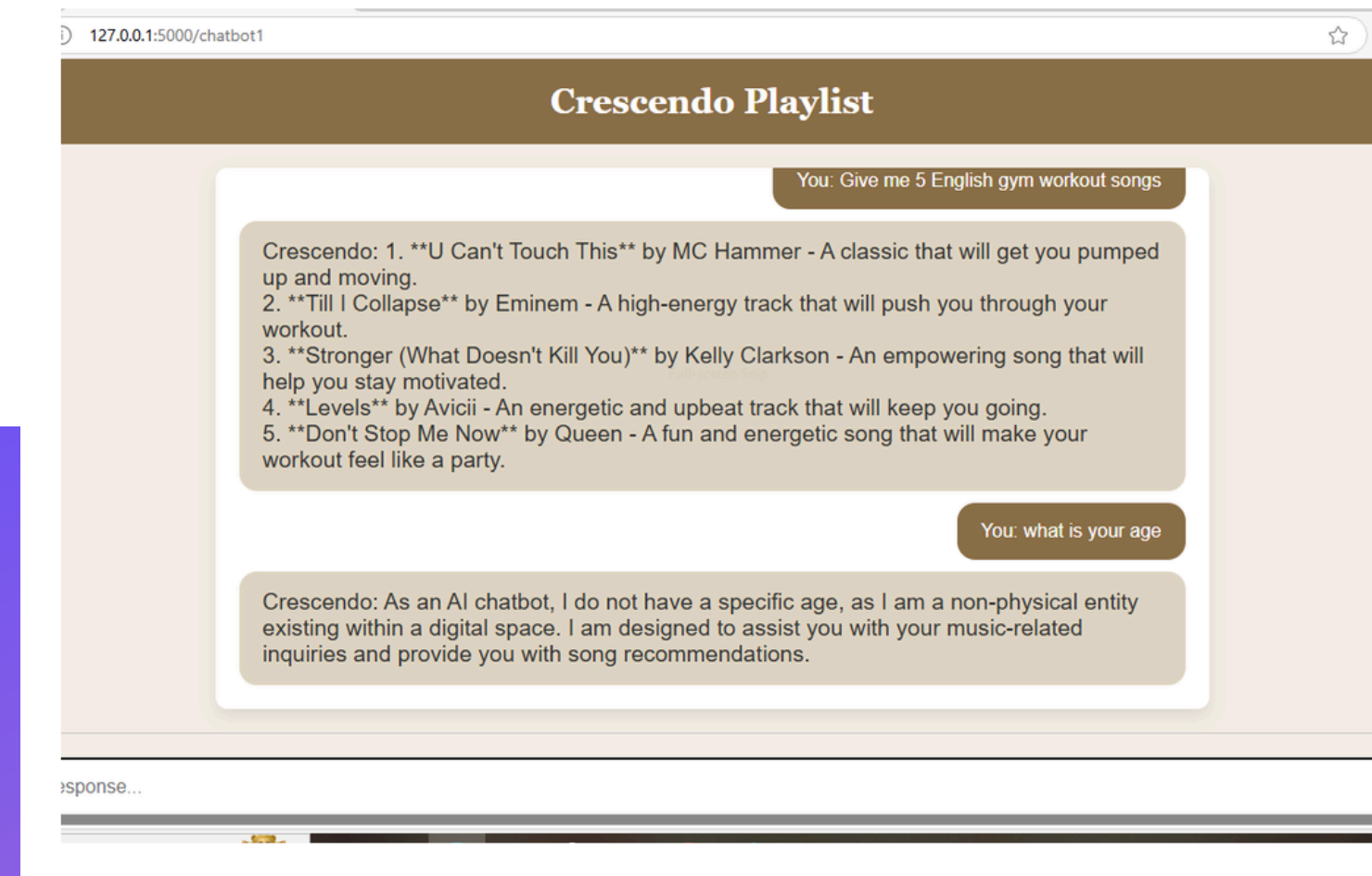


CRESCENDO PLAYLIST CHATBOT

You can create your own:

- **Activity-Based Playlists:** Curated music for activities like gym workouts, dance parties, weddings, and DJ events.
- **Singer-Based Playlists:** Enjoy music centered around your favorite singers or vocalists.
- **Composer-Based Playlists:** Discover tracks based on your preferred music composers.
- **Language-Based Playlists:** Select songs in the language of your choice, from global to regional.
- **Beat-Based Playlists:** Choose music with specific tempos or beats to match your vibe.

You can customize the number of songs and select any combination of these parameters to receive a unique, handpicked music bouquet tailored to your preferences.



CRESCENDO MOOD PLAYLIST CHATBOT

Mood Playlist Chatbot Key Features:

- **Emotional Support & Therapy:** Provides comfort and therapeutic support by engaging with users about their emotions and feelings.
- **Open Up & Express:** Encourages users to share the reasons behind their sadness or emotions, creating a safe space to open up.
- **Feelings Recognition:** Analyzes users' emotions through a thoughtful series of questions, understanding their mood accurately.
- **Tailored Song Suggestions:** Recommends songs based on the user's current mood, providing musical therapy and emotional upliftment.



TECHNICAL APPROACH

Technologies:

Frontend



HTML CSS JAVASCRIPT

Frameworks



BOOTSTRAP Flask

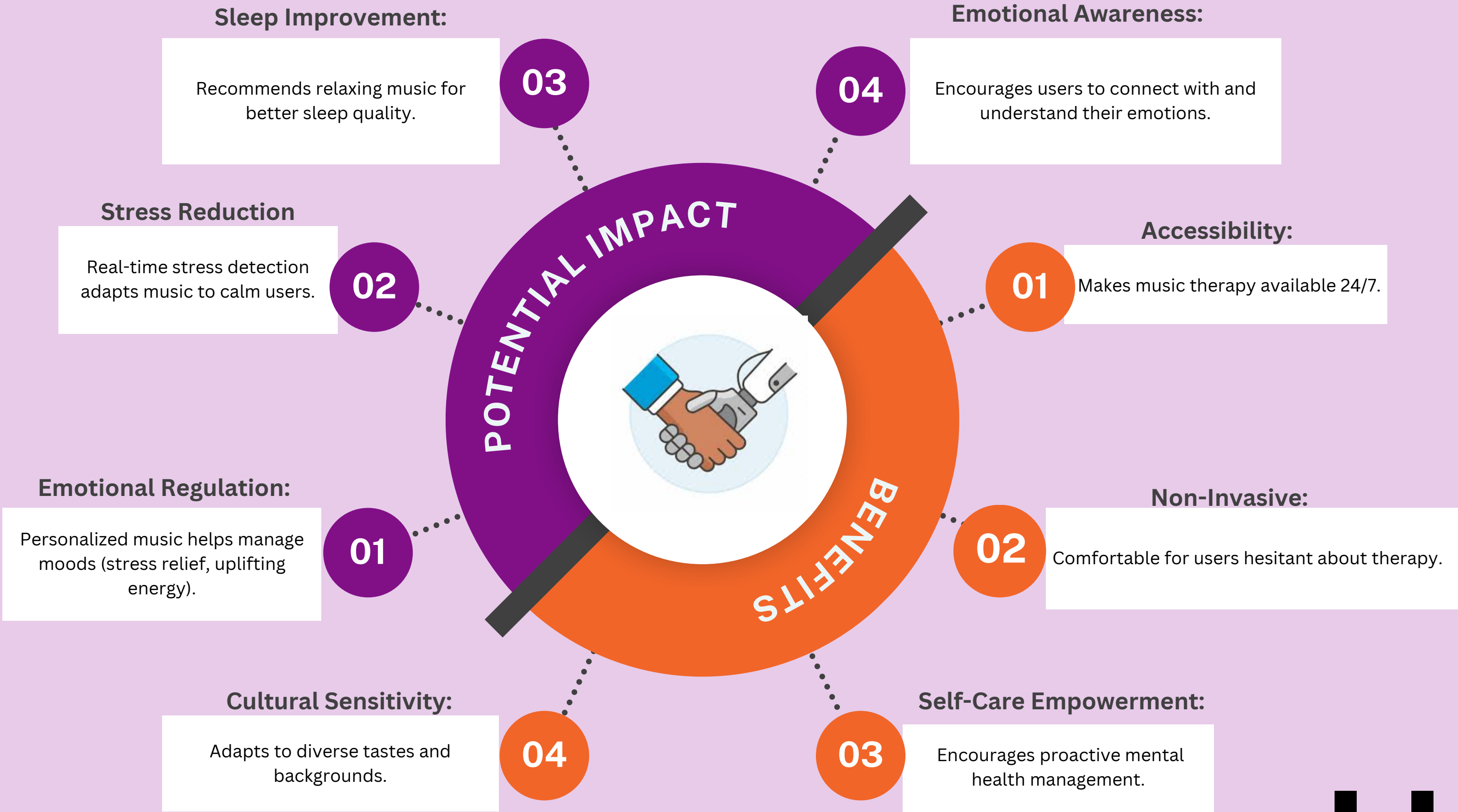
Backend



python™ GitHub

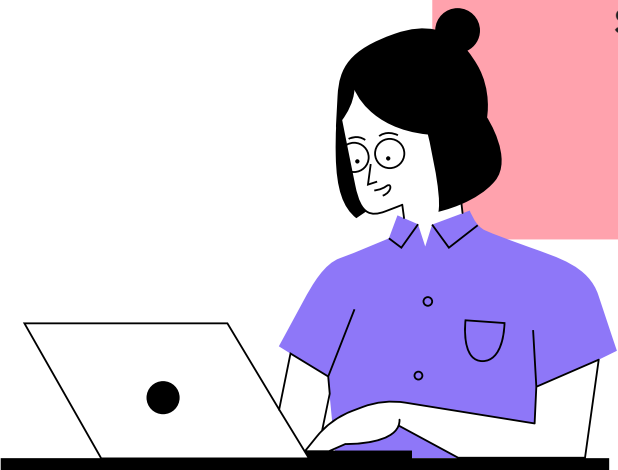
Technology/Library	Purpose
Flask	Backend web framework
Pandas	Data processing and manipulation
NLTK	NLP tasks for tokenization and stemming
FuzzyWuzzy	String matching for recommendations
Google Generative AI	Generating intelligent responses
TfidfVectorizer	Text similarity for recommendations
LLM (Large Language Model)	Google Generative AI (Gemini)

IMPACTS AND BENEFITS



FEASIBILITY

Marketing feasibility: The platform would attract music lovers who are looking for a service that helps them easily find playlists for every mood or occasion. A targeted marketing campaign could attract users by offering solutions for road trips, workout sessions, or moments of relaxation.



Operational Feasibility:

The platform is operationally feasible due to its ability to seamlessly integrate advanced AI-driven emotional support with music therapy. Users can express their emotions without judgment, leveraging the chatbot's empathetic and context-aware interactions, backed by AI-powered coping mechanisms and personalized music recommendations.



Economic feasibility: for the playlist website is high, especially with a freemium or subscription-based revenue model. Users can access basic playlist generation for free, but for advanced features e.g., premium playlists, higher customization, or ad-free experience), they could pay a subscription fee.





Future Implementation

INTEGRATING WITH SMART WATCHES

Building a feature that integrates smartwatch stress detection data and recommends songs based on stress levels, mood, and heart rate

- **Smartwatch Integration:** Uses APIs to fetch stress and heart rate data.
- **Mood Analysis Module:** Use machine learning models to classify moods.

User logs into the website and connects their smartwatch. Website fetches stress data from the smartwatch API. Machine learning processes this data and classifies mood. Backend sends song recommendations to the frontend. User sees their mood and plays recommended songs.





REFERENCES

